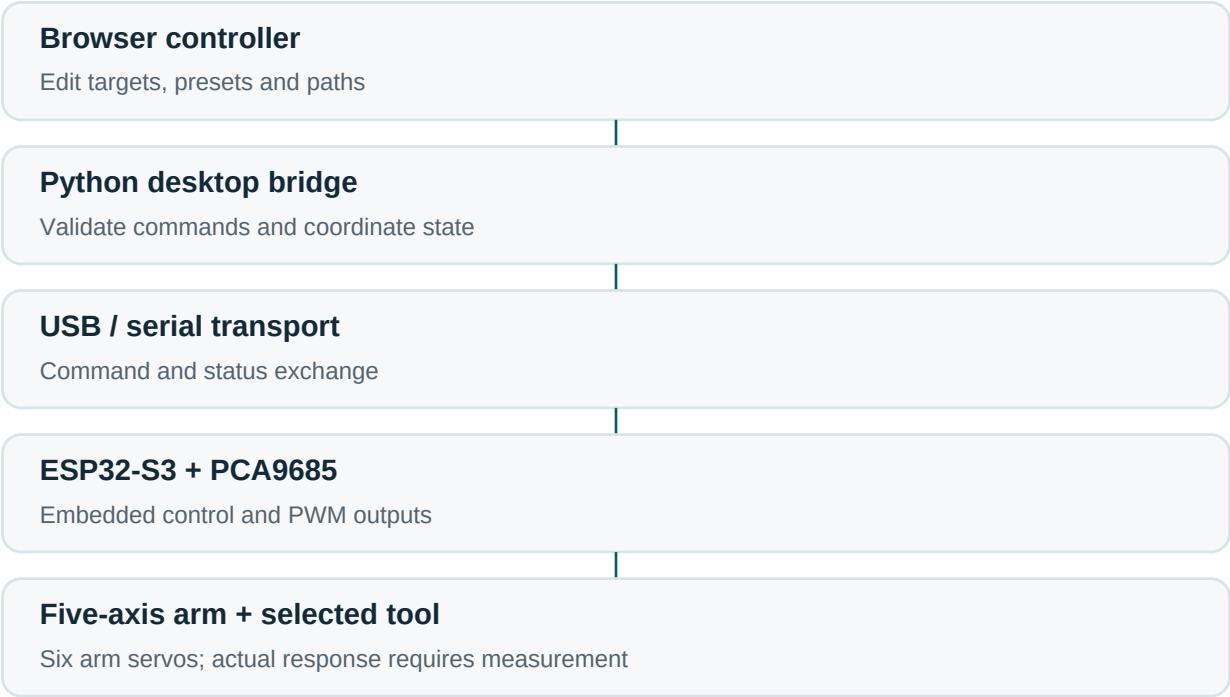


From intent to commanded motion

The browser, desktop bridge and embedded controller form a layered command path. This is an explanatory architecture diagram, not a wiring or deployment guide.



Kinematics. The project began with MATLAB/Python forward and inverse kinematic work. Current browser workflows support Cartesian targeting, joint motion and orientation-aware travel.

State. The current commanded joint pose seeds replay entry. Model and commanded coordinates are not measured physical feedback; backlash, elastic deflection and servo response can create differences.

Tooling. Each attached module needs its own measured mass, tool-centre location and allowed motion/load envelope. A nominal CAD TCP is a starting reference.

AI-assisted workflows supported the creation of this project.

Motion that can be inspected

Named presets store independent waypoint sets. Automatic travel planning and editable paths help structure movements. The main movement-speed control also governs teach/replay.

Recorded update	Result / meaning
Replay interpolation	Synchronized cubic joint interpolation; quintic start/stop easing; fresh approach from current commanded pose.
Persistence and editing	Named presets and location/waypoint persistence; edits preserve existing recordings.
Shoulder guard	Reported back stop near -29.87 degrees informed a -28 degree command limit.
Regression record	28 Python checks plus JavaScript replay/control suites passed in the September 9 source record.
Physical limitations	Command limits are not measured motor torque/current/temperature sensing, and do not guarantee collision-free motion.

Tool revisions are checked digitally

Gripper Rev B: native travel, tooth-section and mesh/slicing checks. Screwdriver R2: native motion, rotation transforms, dense tooth-section checks, archive/exchange comparisons and actual geometry slicing.

The underlying engineering records remain private. The public tooling review summarizes revision-specific results without exposing fabrication files or control source.

What the demonstration establishes

The physical video shows assembly and coordinated movement. It does not establish positioning accuracy, repeatability, payload, grip force, screwdriver torque, electrical fault tolerance or endurance.

Measurement	Purpose
As-built audit	Identify actual servos, printed revision, wiring and tool configuration.
Tool fit and mass	Confirm engagement, fastener access, retention, TCP and load assumptions.
Repeated approaches	Measure spread, bias, backlash and drift across directions and selected loads.
Grip / drive trials	Qualify retention, bit runout, torque, current, temperature and duty off-arm first.
Reliability capture	Correlate intermittent joint response with branch power, signal and serial events.
Whole-arm coverage	Validate physical clearances and reachable paths through the intended pose/load envelope.

Current status

Built and active. Control refinement and tool qualification continue. Intermittent joint/serial behavior remains unresolved; observed recovery and software repairs alone do not establish a root cause.

Public evidence is limited to recorded prototype footage, software regressions and revision-specific CAD checks. New physical measurements will be reported separately.

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